



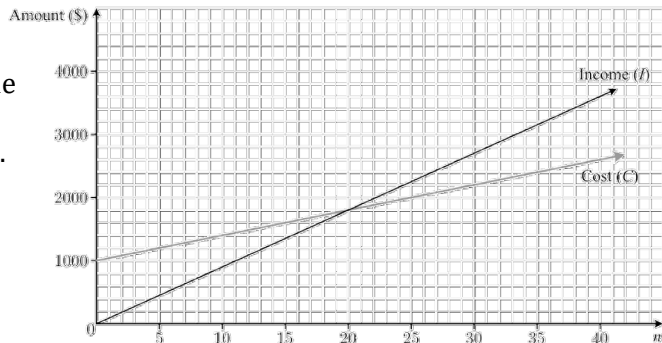
Simultaneous Linear Equation

Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 Caitlin plans to set up a furniture business to sell bedside tables. Below are the income (I) and cost (C) functions for her business.



- a. Write the cost function in the form $C = mn + c$ for the cost of making n tables.

$$m = \frac{\text{rise}}{\text{run}} = \frac{800}{20} = 40 \quad \checkmark \quad c = 1000$$

$$\therefore C = 40n + 1000 \quad \checkmark$$

- b. For how much does Caitlin plan to sell each bedside table?

$$\text{gradient of income graph} = \frac{\text{rise}}{\text{run}} = \frac{1800}{20} = 90$$

$$\therefore I = 90n$$

$$\therefore \text{price of each table} = \$90 \quad \checkmark$$

- c. If Caitlin sells 30 bedside tables, how much profit or loss will she incur?

$$C = 40 \times 30 + 1000 = \$2200 \quad \checkmark$$

$$I = 90 \times 30 = \$2700$$

$$\text{Profit} = 2700 - 2200$$

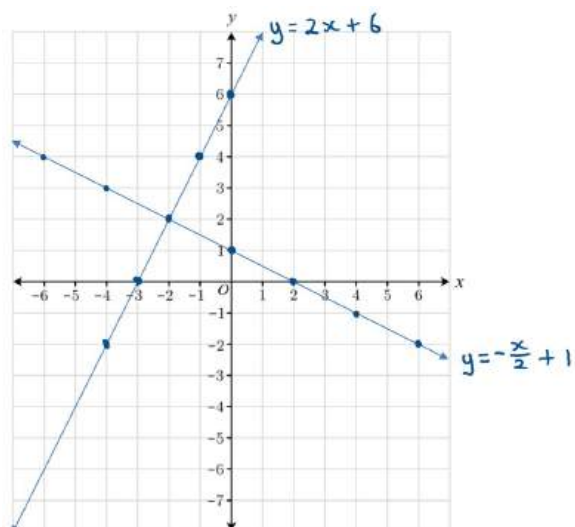
$$= \$500 \quad \checkmark$$

- 2 On the number plane below, graph the equations $y = -\frac{x}{2} + 1$ and $y = 2x + 6$, and hence, solve the equations simultaneously.

Lines are graphed by substituting selected x values into each equation.

Point of intersection of lines = $(-2, 2)$

Solution: $x = -2, y = 2$



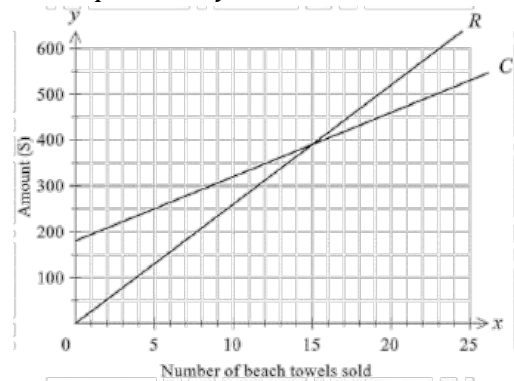
- 3 Anna sells bracelets for \$15 each at a market. Each bracelet cost her \$6.25 to make and she pays a one-off fee of \$70 to sell these at the market.

a. Write an equation to describe the relationship between income (I) and the number of bracelets (n). 1
 Let the income for producing x bubble makers be \$ I
 $I = 15n$

b. Write an equation to describe the relationship between costs (C) and the number of bracelets (n). 1
 Let the costs for producing n bubble makers be \$ C
 $C = 6.25n + 70$

c. How many bubble makers must Anna sell to break-even? 2
 Break-even point (income equals costs)
 $15n = 6.25n + 70$
 $8.75n = 70$
 $n = \frac{70}{8.75} = 8$
 Ava needs to sell 8 bracelets to make a profit.

- 4 A business manufactures and sells beach towels. The graphs representing the costs (C) and revenue (R) are displayed on the same axes below. The cost equation is $y = 14x + 180$. The revenue equation is $y = 26x$.



a. How many beach towels need to be sold to break even? 1

15 beach towels

b. Using the equations, determine the profit/loss amount when 8 beach towels are sold. 2

$$\text{Loss amount} = 14(8) + 180 - 26(8) = \$84$$

- 5 Solve the following pair of equations systematically. 2

$$2x - 3y = 3$$

$$y = 2x - 5$$

$$2x - 3(2x - 5) = 3$$

$$2x - 6x + 15 = 3$$

$$-4x = -12$$

$$x = 3$$

$$y = 2x - 5$$

$$y = 1$$

- 6 Domenica is a furniture maker who builds and sells wooden coffee tables. She has a fixed cost of \$360 to cover the equipment that she uses, and the materials for one table cost her \$180. She sells each table for \$200.

$$C = 360 + 180n$$

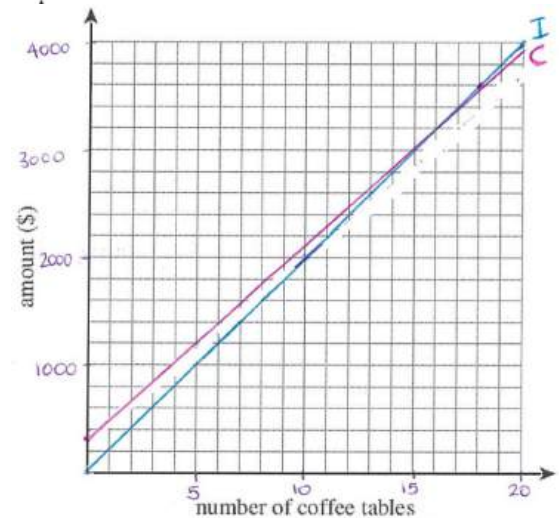
- a. The income from selling the coffee tables (I)

can be modelled by the equation $I = 200n$.

Write down the equation for the cost (C)

associated with the number of coffee tables made (n).

- b. On the grid below, draw the graphs of both equations from part a).



- c. State an estimation for the coordinates of the point of intersection of the two lines drawn in part b). Briefly explain the significance of this point of intersection.

Approx. (18, 3600)

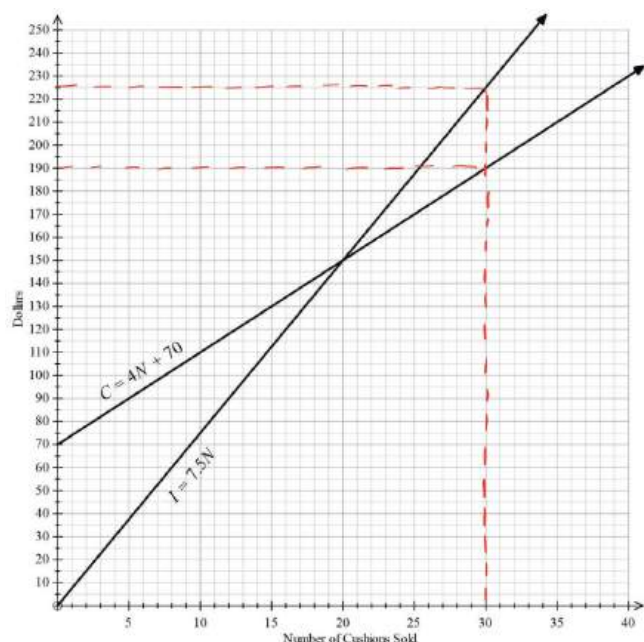
If she ~~Atk~~ sells 10 coffee tables, she will make her money back. More than 10 tables and she starts earning a profit.

- d. Calculate how much profit Domenica will earn when she makes 30 coffee tables.

$$\begin{aligned} C &= 360 + 180(30) = 5760 & \text{Profit} &= \$240 \\ I &= 200 \times 30 = 6000 \end{aligned}$$

- 7 An online retailer of cushions draws the graph below to analyse sales. The lines representing the equations for daily cost (C) and daily income (I) are shown. What is the result on a day where 30 cushions were sold?

A profit of \$35.00



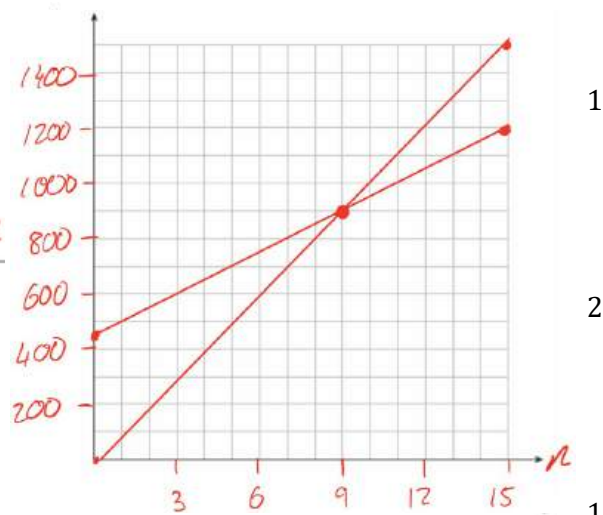
$$225 - 190 = 35$$

- 8 A business has initial daily running costs of \$450 and \$50 per item it produces. The business sells these items for \$100.

- a. Construct equations for I (income) and C (costs).

$$I = 100n \quad C = 450 + 50n$$

- b. Graph the equations onto the plane below for positive values of x .



- c. How many items need to be sold for the business to break even? 9

- 9 The distance travelled by a bicycle varies directly with the number of revolutions made by the pedals. Form a variation equation and find the constant of variation given that the bicycle travels 55 metres for 20 revolutions of the pedals.

$$\begin{aligned} D &= kr \\ 55 &= k(20) \\ k &= \frac{55}{20} = 2.75 \end{aligned}$$



Simultaneous Linear Equation

Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 When Adam joined a sales company he was offered a choice of two different salary contracts.

The details of the contracts are outlined in the table below.

	Salary	End of year commission on total sales
Contract A	\$65 000	2%
Contract B	\$60 000	3%

- a. Find how much Adam would earn under each contract in a year if his total sales were

\$400 000.

$$\text{Contract A} = \$65\,000 + (\$400\,000 \times 2\%) = \$73\,000$$

$$\text{Contract B} = \$60\,000 + (\$400\,000 \times 3\%) = \$72\,000$$

- b. Another employee, Lottie, earned \$90 000 in a particular year. She is on Contract A.

Find her total sales for that year.

Contract A:

$$\$90\,000 = \$65\,000 + (\text{sales} \times 2\%)$$

$$\$90\,000 - \$65\,000 = \text{sales} \times 2\%$$

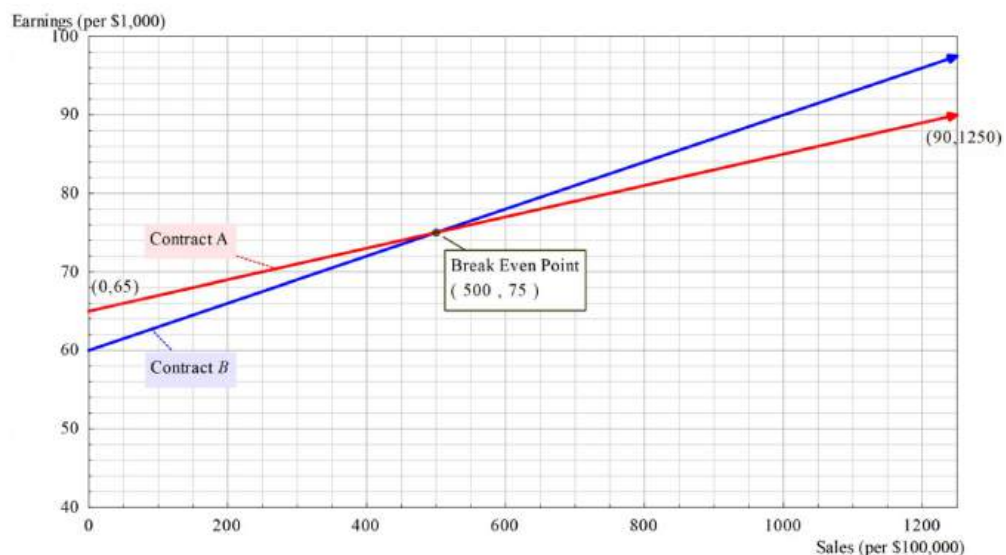
$$\frac{\$90\,000 - \$65\,000}{0.02} = \text{sales}$$

$$\text{Sales} = \$1\,250\,000$$

- c. Write a linear equation to model earnings, E , in relation to sales, S , for Contract A.

$$\text{Contract A: } E = 65\,000 + 0.02S$$

- d. On the grid below, draw the graph of the linear relationship for Contract A and label it.



- e. From your graph or otherwise, determine the total sales for which Contract A and Contract B earn the same amount of money.

Sales \$500 000 have equal earnings of \$75 000 under each contract.

- f. Recommend which contract Adam and Lottie should be on, based on your calculations

above. Adam should chose Contract A. As a starting salesperson he will benefit from having the larger retainer.

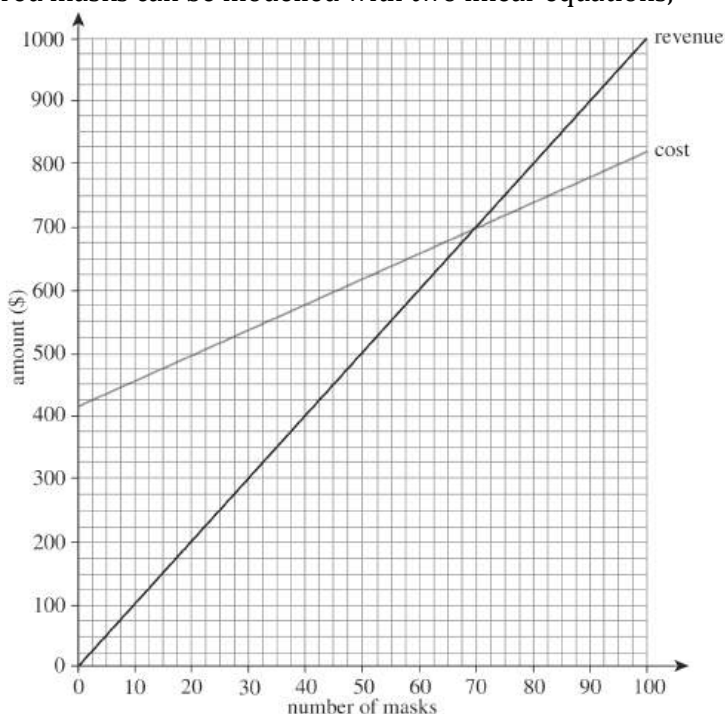
Lottie should move to Contract B. She will earn more with a larger commission from her sales.

- 2 Willow's business Fashion Face Masks makes and sells embroidered face masks. Willow purchases plain cloth masks and embroiders them with different patterns. The cost of each plain mask is \$4 and Willow sells the embroidered masks for \$10 each. Willow invests \$420 in an embroidery machine and a thread kit to make 100 masks. The production cost and revenue generated from the sale of the embroidered masks can be modelled with two linear equations, as shown in the graph.

- a. Using the graph, solve the two equations simultaneously and explain the significance of this solution for Willow's business.

Reading from the graph, the solution is (70, 700). This point of intersection is the break-even point where the revenue and cost of Willow's business are equal.

- b. How many masks does Willow need to sell to obtain a revenue of \$350?



Reading from the graph, Willow would need to sell 35 masks.

Alternatively, where R is revenue and N is the number of masks:

$$R = 10N \quad N = \frac{R}{10} = \frac{350}{10} = 35$$

- c. How much profit does Willow make if she sells 100 masks?

Where R is revenue, N is the number of masks, and C is cost:

- $R = 10N$

- $C = 420 + 4N$

For 100 masks:

$$R = 10 \times 100 = \$1000$$

$$C = 420 + 4 \times 100 = \$820$$

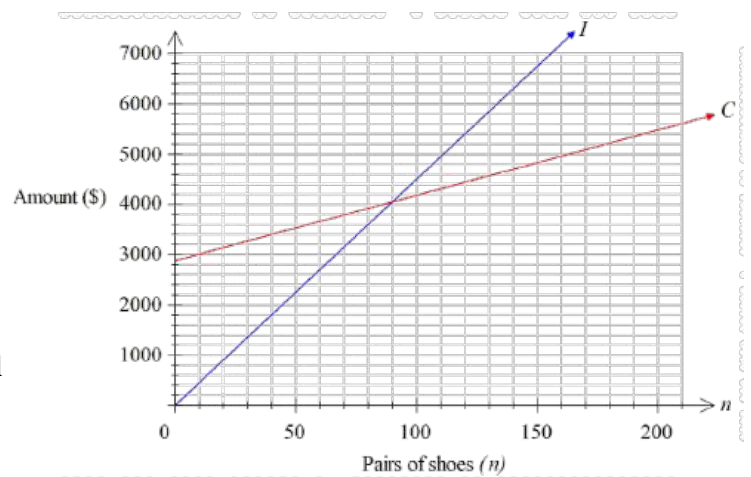
- 3 Last Saturday, Luke had 165 followers on social media. Rhys had 537 followers. On average, Luke gains another 3 followers per day and Rhys loses 2 followers per day.

If x represents the number of days since last Saturday and y represents the number of followers, write the pair of equations to model this situation.

Luke: $y = 165 + 3x$

Rhys: $y = 537 - 2x$

- 4 A cobbler repairs the soles of old shoes for \$45 per pair. It costs \$13 for new soles for a pair of shoes and \$2880 for the equipment needed to repair them. The income, I , and cost, C , functions for repairing n pairs of shoes are illustrated in the following graph.



- a. Determine the equations of the cost, C , and income, I , functions for fixing n pairs of shoes. 2

$$C = 13n + 2880 \quad I = 45n$$

- b. Determine the break-even point and explain what it represents. 2

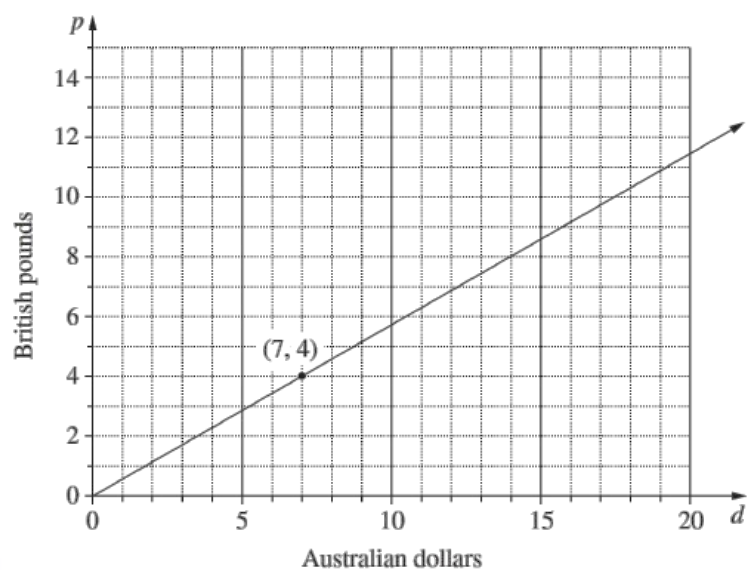
(90, 4050) The cobbler has to fix more than 90 pairs of shoes to make profit.

- c. Write down a formula for the profit, P , made for fixing n pairs of shoes. 1

$$\text{Profit} = \text{Income} - \text{Costs}$$

$$P = 45n - (13n + 2880) = 32n - 2880$$

- 5 The relationship between British pounds (p) and Australian dollars (d) on a particular day is shown in the graph.



- a. Write the direct variation equation relating British pounds to Australian dollars in the form $p = md$. Leave m as a fraction. 1

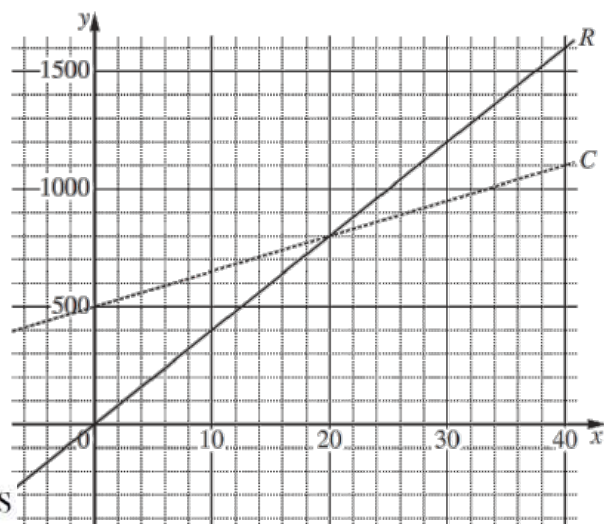
$$p = \frac{4}{7}d$$

- b. The relationship between Japanese yen (y) and Australian dollars (d) on the same day is given by the equation $y = 76d$. Convert 93 100 Japanese yen to British pounds. 2

$$d = \frac{y}{76} = \frac{93100}{76} = 1225, \quad p = \frac{4}{7} \times 1225 = 700$$

- 6 A small business makes and sells bird houses.

Technology was used to draw straight-line graphs to represent the cost of making bird houses (C) and the revenue from selling bird houses (R). The x -axis displays the number of bird houses and the y -axis displays the cost/revenue in dollars.



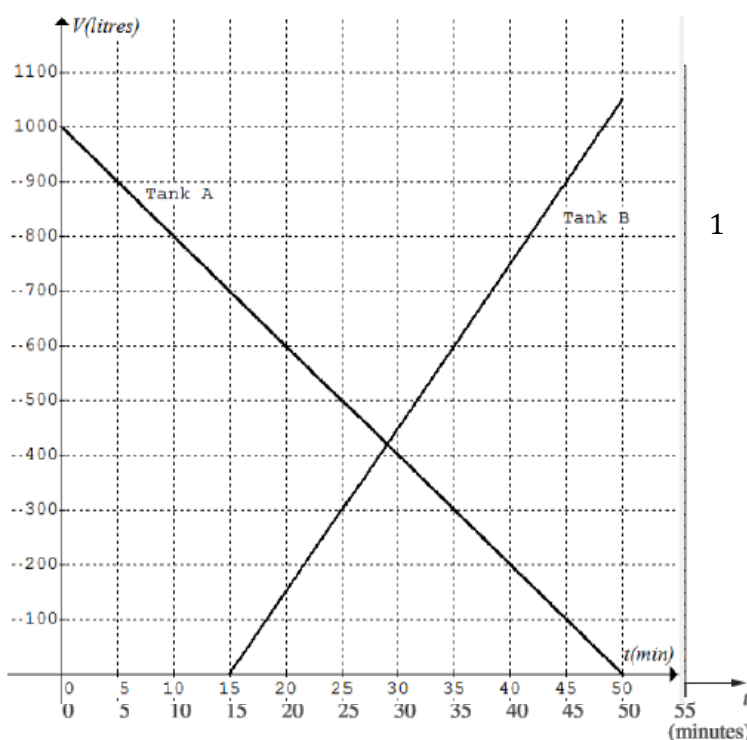
- a. How many bird houses need to be sold to break even? **20 bird houses**
- b. By first forming equations for cost (C) and revenue (R), determine how many bird houses need to be sold to earn a profit of \$1900.

$$C = 15x + 500, R = 40x$$

$$\text{Profit } (\$) = R - C = 25x - 500 = 1900, x = 96$$

- 7 There are two tanks on a property, Tank A and Tank B. Initially, Tank A holds 1000 litres of water and Tank B is empty.

- a. Tank A begins to lose water at a constant rate of 20 litres per minute. The volume of water in Tank A is modelled by $V = 1000 - 20t$, where V is the volume in litres and t is the time in minutes from when the tank begins to lose water. On the grid



- below, draw the graph of this model and label it as Tank A.
- b. Tank B remains empty until $t = 15$ when water is added to it at a constant rate of 30 litres per minute. By drawing a line on the grid on the previous page, or otherwise, find the value of t when the two tanks contain the same volume of water.

$$\text{Tank A: } V = 1000 - 20t; \text{ Tank B: } V = 30(t - 15)$$

$$\text{Same volume: } 30(t - 15) = 1000 - 20t, 50t = 1450, t = 29$$

- c. Using the graphs drawn, or otherwise, find the value of t (where $t > 0$) when the total volume of water in the two tanks is 1000 litres.

$$30(t - 15) + 1000 - 20t = 1000, t > 0 \therefore 10t = 450, t = 45$$